Java Code

```java
package testpet;

/**
 *
 * @author humai
 */
public class TestPet {

    /**
     * @param args the command line arguments
     */
    public static void main(String[] args) {
        Pet myPet = new Pet("Charlie");

        System.out.println(myPet.toString());
        System.out.println(myPet.speak());
        System.out.println(myPet.move());

        System.out.println("\n--- Testing Cat ---");

        Cat myCat = new Cat("Zeus");
        System.out.println(myCat.toString());
        System.out.println(myCat.speak());
        System.out.println(myCat.move());

        System.out.println("\n--- Testing Dog ---");

        Dog myDog = new Dog("Rex",3);
        System.out.println(myDog.toString());
        System.out.println(myDog.speak());
        System.out.println(myDog.move());
        System.out.println(myDog.howDangerous());

        Pet[] pets = new Pet[50];

        pets[0] = new Dog("Bolt", 1);
        pets[1] = new Dog("Spike", 2);
        pets[2] = new Dog("Ghost", 3);

        pets[3] = new Cat("Gizmo");
        pets[4] = new Cat("Zeus");
        pets[5] = new Cat("Honey");


        for (int i = 6; i < pets.length; i++){
            pets[i] = new Pet("Pet" + i);
        }

        System.out.println("\n--- Loop: All Pets ---");

```

```java
        for (int i = 0; i < pets.length; i++){
            System.out.println("----------------------------");
            System.out.println(pets[i].toString());
            System.out.println(pets[i].speak());
            System.out.println(pets[i].move());
        }


        System.out.println("\n--- Loop: Dangerous Dogs Only ---");

        for (int i = 0; 1 < pets.length; i++){
            if (pets[i] instanceof Dog){
                Dog dog = (Dog) pets[i];

                if (dog.getDangerLevel() >= 2){
                    System.out.println(dog.toString() + " - " + dog.howDangerous());
                }
            }
        }


    }

}

```

Java Code

```java
package testpet;

/**
 *
 * @author humai
 */
public class Pet {
    private String name;

    public Pet(){
        this.name = "No Name";
    }

    public Pet(String name){
        this.name = name;
    }

    public String getName(){
        return name;
    }

    public void setName(String name){
        this.name = name;
    }

    public String toString(){
        return "My name is " + name;
    }

    public String speak(){
        return "I am your cute pet.";
    }

    public String move(){
        return "I cannot move as I do not know what pet I am.";
    }


}
```

Java Code

```java
package testpet;

/**
 *
 * @author humai
 */
public class Dog extends Pet{
    private int dangerLevel;

    public Dog(){
        super("NoNameDog");
        this.dangerLevel = 1;
    }

    public Dog(String name, int dangerLevel){
        super(name);
        setDangerLevle(dangerLevel);
    }

    public int getDangerLevel(){
        return dangerLevel;
    }

    public void setDangerLevle(int dangerLevel){
        if (dangerLevel == 1 || dangerLevel == 2 || dangerLevel == 3){
            this.dangerLevel = dangerLevel;
         }
        else{
            this.dangerLevel = 1;
        }
    }

    @Override
    public String speak(){
        return "Dogs say Woof-woof";
    }

    @Override
    public String move(){
        return "Dog walk and run. Wish I could climb a tree to catch a cat.";
    }

    public String howDangerous(){
        switch (dangerLevel){
            case 1:
                return "Suitable for children.";
            case 2:
                return "Need a strong owner that can control me.";
            case 3:
                return "Watch-dog only, as I attack and bite strangers";
```

```
            default:
                return "Unknown danger level";
        }
    }
}

```

Java Code

```java
package testpet;

/**
 *
 * @author humai
 */
public class Cat extends Pet {

    public Cat(){
        super("NoNameCat");
    }

    public Cat(String name){
        super(name);
    }

    @Override
    public String speak(){
       return "Cats say Miaauw";
    }

    @Override
    public String move(){
        return "Cat walk, jump and climb";
    }

}

```